

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-0917

Eagleville 138 kV
September 18, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0917. The Interconnection Customer proposes a 492.4 MW natural gas combined cycle facility interconnecting at the Eagleville 138 kV substation in the Virginia area. The request was received on January 12, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 301 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$35,577,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
EAGLEVILLE - INDEPENDENT HILL D.P 138 kV L	MEDICAL - EAGLEVILLE 138 kV	215.7	224	100.0
INDEPENDENT HILL D.P - INNOVATION 138 kV	MEDICAL - EAGLEVILLE 138 kV	214.2	216	100.0
MEDICAL - EAGLEVILLE 138 kV	INDEPENDENT HILL D.P - EAGLEVILLE 138 kV L	213.7	226	100.0
DUMFRIES - MEDICAL 138 kV	INDEPENDENT HILL D.P - EAGLEVILLE 138 kV	186.1	239	100.0
DMFRIES - HEISKELL 138 kV	INDEPENDENT HILL D.P - EAGLEVILLE 138 kV	164.6	257	100.0
HEISKELL - POSSUM PT 138 kV	EAGLEVILLE - INDEPENDENT HILL D.P 138 kV	140.9	368	85.5
BRODHEAD SW STA - CS MURPHY POINT LLC 138 kV	POSSUM PT - HEISKELL 138 kV	116.1	262	42.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
EAGLEVILLE - INDEPENDENT HILL D.P 138 kV L	Rebuild line with higher-capacity conductor	\$12,203,000	100% (\$12,203,000)
INDEPENDENT HILL D.P - INNOVATION 138 kV	Rebuild line with higher-capacity conductor	\$2,921,000	100% (\$2,921,000)
MEDICAL - EAGLEVILLE 138 kV	Rebuild line with higher-capacity conductor	\$9,261,000	100% (\$9,261,000)
DUMFRIES - MEDICAL 138 kV	Rebuild line with higher-capacity conductor	\$3,964,000	100% (\$3,964,000)
DUMFRIES - HEISKELL 138 kV Line	Rebuild line with higher-capacity conductor	\$4,533,000	100% (\$4,533,000)
HEISKELL - POSSUM PT 138 kV	Rebuild line with higher-capacity conductor	\$826,000	100% (\$826,000)
BRODHEAD SW STA - CS MURPHY POINT LLC 138 kV	Rebuild line with higher-capacity conductor	\$1,930,000	97% (\$1,869,000)
	Total allocated to project		\$35,577,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.